

Page 1 - Spring Boot

Spring Boot is a Java framework for building production-ready applications.

Features include auto-configuration, starter dependencies, and embedded servers.

It simplifies enterprise application development.

Page 2 - Dependency Injection

Dependency Injection reduces coupling between classes.

The Spring IoC container manages object creation and wiring.

It improves testability and maintainability.

Page 3 - Microservices

Microservices break an application into independently deployable services.

Each service owns its data and communicates over APIs.

They improve scalability and resilience.

Page 4 - RAG

Retrieval-Augmented Generation combines document retrieval with an LLM.

Documents are chunked, embedded, stored in a vector database, and retrieved.

Relevant chunks are provided to the LLM as context.

Page 5 - Embeddings & Vector DB

Embedding models convert text into vectors.

Vector databases such as pgvector perform similarity search.

Ollama can use llama3.2 for chat and nomic-embed-text for embeddings.